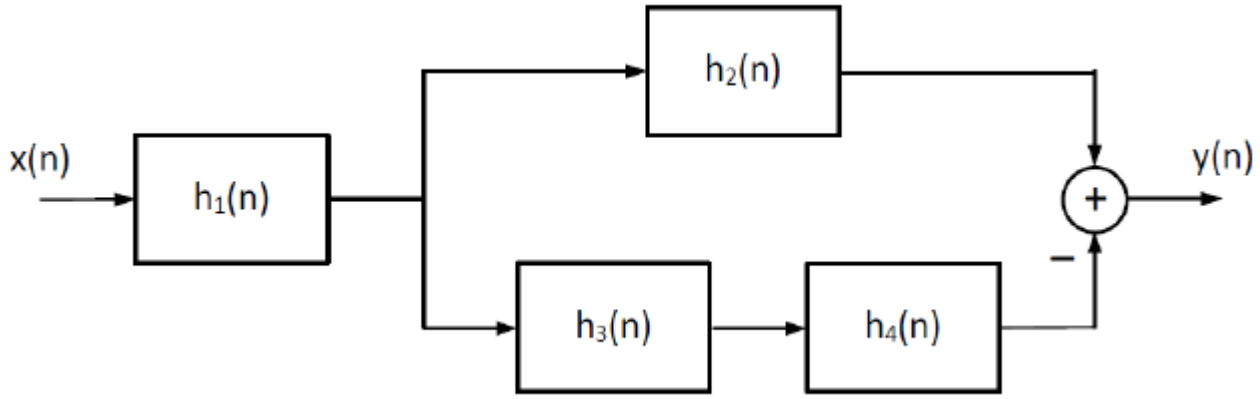
# LCCDE and convolution

Practice questions · Day 04 · Tue 27 Oct · pack D2

## 5. Practice

### From your course files

**T2 Q4 (tutorial).** Consider the interconnection of LTI systems shown below.
(a) Express the overall impulse response of the system $h(n)$ in terms of $h_1(n)$, $h_2(n)$, $h_3(n)$ and $h_4(n)$.
(b) Determine $h(n)$ when $h_1(n) = \{\underset{\uparrow}{2}, 1, 2\}$, $h_2(n) = h_3(n) = (n+1)u(n)$, and $h_4(n) = \delta(n-2)$. Is this system (i) causal and (ii) stable?




T2 Q4 figure (from the tutorial sheet)

**T3 Q2 (tutorial).** A LSI system is described by the LCCDE

$$y[n] = \frac{1}{4}\big(x[n] + 2x[n-1] + x[n-2]\big).$$

(a) Write down the characteristic modes of this system.
(b) Find the output of the system if the input signal is $x[n] = 5^{-n}u[n]$.

**T3 Q4 (tutorial).** A LSI system is described by the LCCDE

$$y[n] + y[n-1] + \frac{1}{4}y[n-2] = x[n].$$

The initial conditions are $y[-2] = 2$, $y[-1] = -1$.
(a) Write down the characteristic modes of this system.
(b) Find the output of the system if the input signal is $x[n] = 2^{-n}u[n]$.

**T3 Q5 (tutorial).** Determine the impulse response $h[n]$ of the following DT LSI systems:
(a) $y[n] = x[n] + 2x[n-1] + 3x[n-2] + 4x[n-4]$
(b) $y[n] + 0.2y[n-1] - 0.08y[n-2] = x[n]$
(c) $y[n] - 2y[n-1] + y[n-2] = x[n]$

**T3 Q6 (tutorial).** Compute the DT convolution $y[n] = h[n] * x[n]$ for the following signals:
(a) $x[n] = \{\underline{-1}\ \ 0\ \ 1\ \ 3\}$, $h[n] = \{-2\ \ 3\ \ \underline{1}\ \ 2\ \ 4\}$ (the underline marks $n = 0$)
(b) $x[n] = 3^n u[n]$, $h[n] = 2^{-n}u[n]$.
(c) $x[n] = \exp(j\omega_0 n)u[n]$, $h[n] = 2^{-n}u[n]$, where $\omega_0$ is a constant.
(d) $x[n] = \cos(\omega_0 n)u[n]$, $h[n] = 2^{-n}u[n]$, where $\omega_0$ is a constant.

## New questions

**Q1 (easy) [2].** $x(n) = \{1, \underset{\uparrow}{2}, -1\}$ and $h(n) = \{\underset{\uparrow}{3}, 0, 1, 2\}$. Find $y(n) = x(n) * h(n)$, marking $n = 0$.

**Q2 [3].** Find $y(n) = x(n) * h(n)$ in closed form for $x(n) = u(n) - u(n-4)$ and $h(n) = (0.5)^n u(n)$.

**Q3 [3].** $y(n) - 0.5y(n-1) = x(n) - x(n-1)$. Find $h(n)$ and state whether the system is causal, stable, FIR or IIR.

**Q4 [4].** $y(n) - y(n-1) + 0.25y(n-2) = x(n)$, initially at rest, with $x(n) = (1/3)^n u(n)$. Find $y(n)$ in closed form.

**Q5 [3].** $y(n) - y(n-1) + 0.5y(n-2) = x(n)$. (a) Find the characteristic modes. (b) Find $h(n)$ in real form. (c) Is the system stable?

**Q6 (chained) [7].** System $S_1$ is $y(n) = 0.5y(n-1) + x(n)$. System $S_2$ has $h_2(n) = (0.25)^n u(n)$. They are connected in cascade.

a. Write $h_1(n)$. [1]
b. Find the overall $h(n)$ in closed form. [2]
c. Show that $h(n)$ satisfies $y(n) - 0.75y(n-1) + 0.125y(n-2) = x(n)$ by checking $h(0)$ and $h(1)$, and explain why the modes are $0.5^n$ and $0.25^n$. [2]

d. Find the cascade's output for $x(n) = (0.5)^n u(n)$. [2]




Q6: $S_1$ in cascade with $S_2$

## 6. Quick recall check

1. Which coefficients form the characteristic equation, and what are the "modes" of an FIR LCCDE?
2. What homogeneous solution do you use for a double root at $\lambda = -\frac{1}{2}$?
3. Write $a^n u(n) * b^n u(n)$ for $a \neq b$, and for $a = b$.
4. Is $h(n) = \cos(\pi n/3)u(n)$ stable? Why?
5. $x$ has 4 samples starting at $n = -3$ and $h$ has 5 samples starting at $n = 0$. Where does $y$ start, and how long is it?